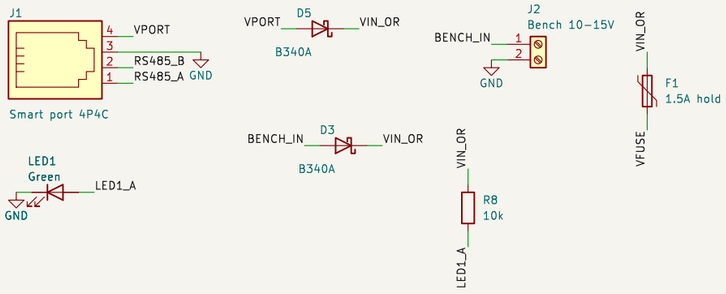
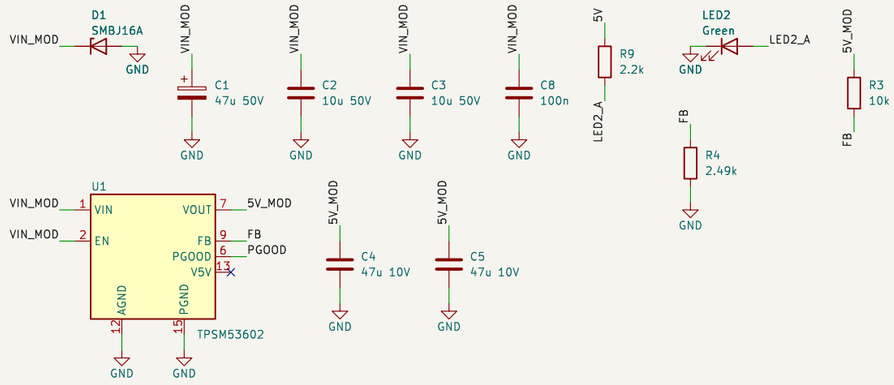
1 POWER INPUT

D5 and D3 OR the two inputs. Either can be plugged in alone or both at once.
Meter the J1 pinout before the first plug-in. A wrong pinout can put battery on GND.



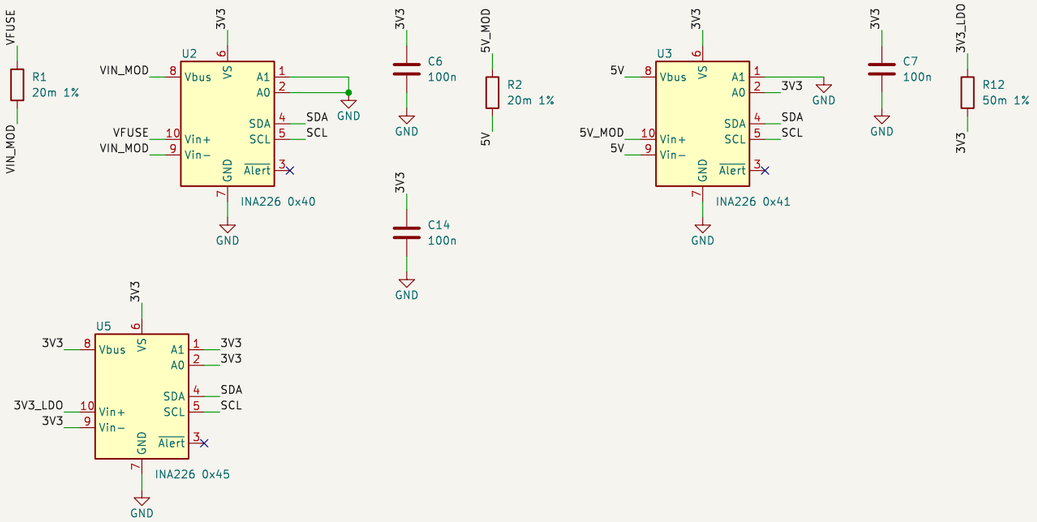
2 5 V REGULATOR

TPSM53602 set to 5.0 V: RFBT 10k, RFBB 2.49k (datasheet Table 7-1). 1.4 MHz fixed.
EN tied to VIN. AGND joins PGND at the module. Input 2 x 10u, output 2 x 47u.



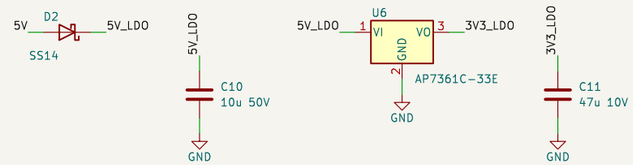
3 CURRENT SENSE

Route IN+ and IN- as Kelvin traces from the inside edge of each shunt pad.
Addresses: U2 0x40, U3 0x41, U5 0x45.



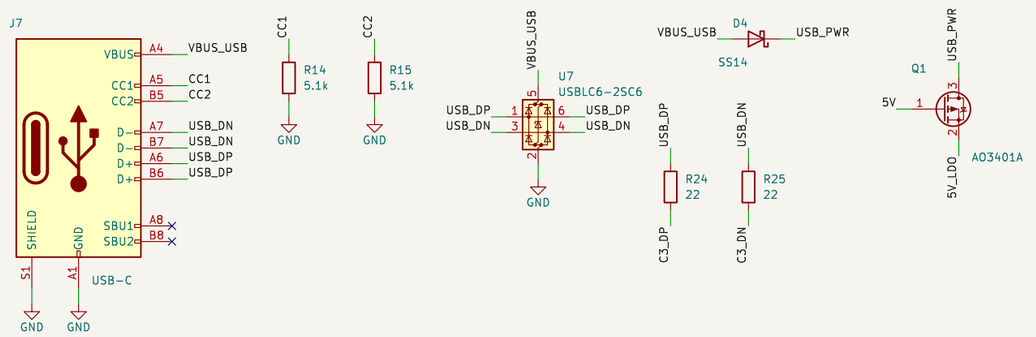
4 3.3 V REGULATOR

5V_LDO is fed by 5V through D2, or by USB through D4 and Q1.
Q1's gate sits on the 5 V rail: robot or bench power on turns Q1 off, so USB only carries data.



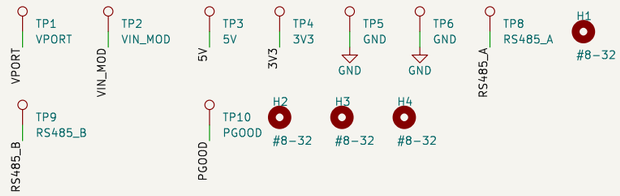
5 USB-C

Separate 5.1k on each CC pin. USB D+/D- reach GPIO19/GPIO18 through 22 ohm R24, R25.

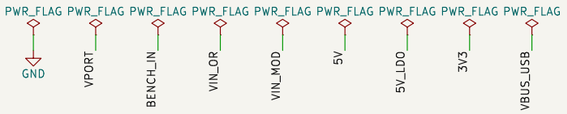


8 TEST POINTS, MECHANICAL

Test points are bare 3.4 x 1.8 mm copper pads, nothing to buy.
Same land as a Keystone 5015, so a loop can be soldered on later.

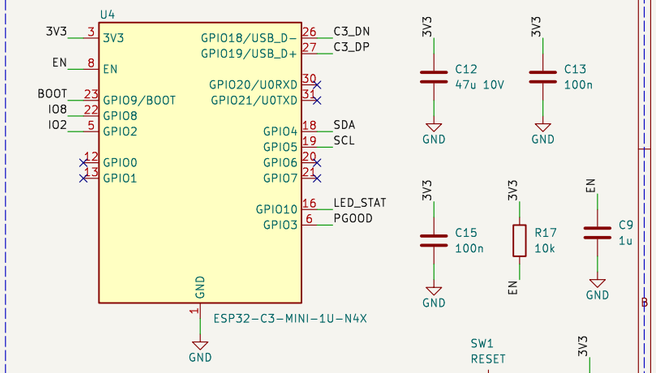


Power flags (tell ERC these nets are driven)

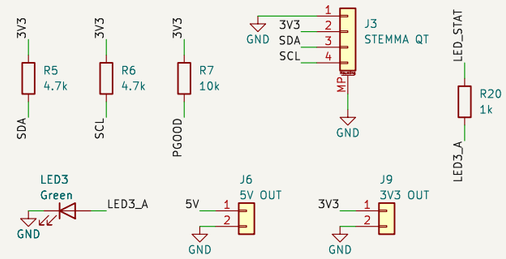


6 ESP32-C3 MCU

USB-C is the built-in USB serial and JTAG port: flash and print over one cable.
Hold BOOT, tap RESET for download mode. R21, R22 hold the GPIO2 and GPIO8 straps high.
UART0 (GPIO20/21) is not broken out. USB-C is the console.



7 I2C, LEDS, OUTPUTS



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Centurion board family
UWAT VEXU, University of Waterloo

Sheet: /
File: powerJackal.kicad_sch

Title: Jackal: smart port power test board

Size: A3	Date: 2026-10-12	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1